Math is Explaining and Sharing — Homework

Lesson 1-4 • Standard 5.MD.C.5

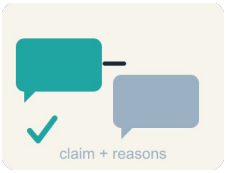
Name: _____ **Period:** _____ **Date:** _____

Today's Goals

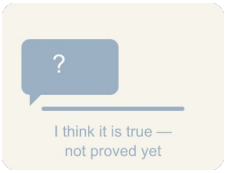
Content: I can construct an argument using equations, drawings, or words, and use volume to defend a recommendation.

Language: I can explain my thinking with clear math vocabulary using the words argument, conjecture, and cubic units.

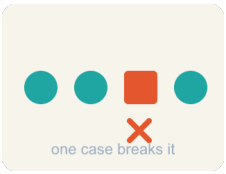
Key Words



argument (argumento) — A chain of reasons — equations, drawings, or words — used to defend your thinking.



conjecture (conjetura) — A mathematical statement you believe is true but have not yet proven.



counterexample (contraejemplo) — A single example that shows a statement is not always true.



volume (volumen) — The amount of space a solid figure takes up, measured in cubic units.



cubic unit (unidad cúbica) — A cube with edges 1 unit long — the unit used to measure volume.

Practice

Show your work. Use the blank space under each problem.

1. An egg muffin is about 60 calories. A plate holds 4 egg muffins. About how many calories are on the plate?

A. About 15
B. About 64
C. About 240
D. About 600

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2. A large cereal box is 8 in long, 3 in wide, and 10 in tall. What is its volume?

A. 21 cubic inches
B. 80 cubic inches
C. 240 cubic inches
D. 240 square inches

3. Three large boxes hold 240 cubic inches each. Two jumbo boxes hold 400 cubic inches each. Which is the correct volume comparison?

A. 240 cubic inches vs. 400 cubic inches — the jumbo option holds more
B. 480 cubic inches vs. 800 cubic inches — the jumbo option holds more
C. 720 cubic inches vs. 800 cubic inches — the jumbo option holds more
D. 720 cubic inches vs. 800 cubic inches — the large option holds more

4. One garden bed is 4 ft long, 3 ft wide, and 2 ft deep. A second bed is 6 ft long, 2 ft wide, and 2 ft deep. How much soil fills BOTH beds?

A. 19 cubic feet
B. 24 cubic feet
C. 48 cubic feet
D. 96 cubic feet

5. The group found the volume by decomposing the garden into prisms and adding. Why would they compute it a SECOND, different way?

A. Because the first way is always wrong
B. To get a bigger number for the soil order
C. If both ways give the same answer, the solution is accurate
D. Because volume changes each time you measure it

6. A student recommends the three large boxes because the cereal stays fresher — each box is open for less time. Is this argument convincing? Explain.

7. Which statement is a CONJECTURE?

A. “8 × 3 × 10 = 240.”
B. “I think the two jumbo boxes hold more cereal than the three large boxes.”
C. “Volume is measured in cubic units.”
D. “Yuzuki bought cereal.”

8. Use precise vocabulary to explain how you would find the soil needed for a bed 5 ft long, 2 ft wide, and 2 ft deep. Include the answer with its unit.

9. Compute the volume of each cereal box.

Box (l × w × h, inches)	Volume (cubic inches)
8 × 3 × 10	
5 × 8 × 10	

Box (l × w × h, inches)	Volume (cubic inches)
6 × 4 × 5	